

CSCI 5454

Design and Analysis of Algorithms

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	AI Engineer (39.2 / 100)	29.0 / 100
SPECIALIZATION On-Campus Elective		

29

CURRENT AVG SCORE
across 6 job roles

76

POTENTIAL SCORE
with recommended additions

+69

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Techniques for algorithm design, analysis of correctness and efficiency; divide and conquer, dynamic programming, probabilistic methods, advanced data structures, graph algorithms, etc. Lower bounds, NP-completeness, intractability.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	39.2	75.1	+35.9	HIGH
ML Engineer	32.8	68.2	+35.4	HIGH
Applied Scientist	31.0	100.0	+69.0	HIGH
Machine Learning Engineer	28.7	77.7	+49.0	HIGH
Data Scientist	22.2	73.4	+51.2	HIGH
MLOps Engineer	19.9	61.0	+41.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

39.2 → 75.1 (+35.9 pts)

perception models learning model systems problems detection machine

ML Engineer

32.8 → 68.2 (+35.4 pts)

discovery learning machine systems models model problems building

Applied Scientist

31.0 → 100.0 (+69.0 pts)

biology discovery scientific computational research interdisciplinary problems models

Machine Learning Engineer

28.7 → 77.7 (+49.0 pts)

sets learning analytics tools solutions exploratory business mobile

Data Scientist

22.2 → 73.4 (+51.2 pts)

definition sets analytics presentation social problems approaches

MLOps Engineer

19.9 → 61.0 (+41.1 pts)

detection model systems learning machine training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

